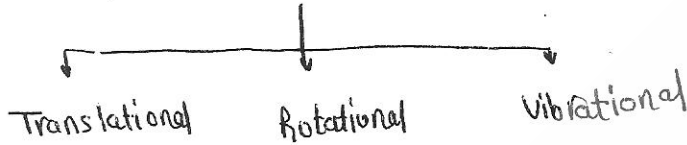


Chapter "2"

2.1) position, velocity and speed:

→ There are [3] Types of motion



→ vector and scalar quantities:

Force

velocity

Mass

work

power

we are interested in
magnitude and
direction

we are interested
in magnitude

→ Position: Location of the particle with respect to reference point.

→ Distance: The long of the path followed by the particle

→ Displacement: Distance but with not ignoring the direction (+, -)

Example: A runner moves from point A to point B Find the Distance and the Displacement.

— Solution: —

$$\Delta x = x_f - x_i = 7 - 2 = 5\text{m}, \rightarrow \text{Displacement}$$

(+)

$$d = 5\text{m}$$



→ velocity:

$$V_{\text{avg}} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}$$

Unit: m/s

2.2) Instantaneous velocity and speed:

$$V_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

⊗ Example: A particle moves along the x-axis it's position = $\frac{1}{\sqrt{2t^2+3}}$

- a) find the displacement in the time interval $t = 1.3$ s
- b) find the average velocity in the time interval $t = 1.3$ s
- c) find the instantaneous velocity at $t = 5$ s

— Solution! —

$$a) \Delta x = x_f - x_i = x(3s) - x(1s)$$

$$= \frac{1}{\sqrt{2(9)+3}} - \frac{1}{\sqrt{2(1)+3}} = -0.229 \text{ m}$$

$$b) V_{x, \text{avg}} = \frac{\Delta x}{\Delta t} = \frac{-0.229}{2} = -0.114 \text{ m/s}$$

$$c) \frac{dx}{dt} = \frac{-1 \times \frac{4t}{2\sqrt{2t^2+3}}}{(2t^2+3)} = \frac{-2t}{(2t^2+3)^{\frac{3}{2}}}$$

$$V_x(t=5s) = \left| \frac{-2 \times 5}{(2 \times 25 + 3)^{\frac{3}{2}}} \right| = 0.026 \text{ m/s}$$

2.3) The particle Under constant velocity:

Constant speed $\Rightarrow$ Instantaneous speed = average speed

2.4) Acceleration: Units $[m/s^2]$

average acceleration

$$a_{\text{avg}} = \frac{\Delta V_x}{\Delta t} = \frac{V_{x,f} - V_{x,i}}{t_f - t_i}$$

instantaneous acceleration

$$a_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta V_x}{\Delta t} = \frac{d^2 x}{dt^2}$$

* NOTE!

I) $v \rightarrow$
 $a \rightarrow$

The particle is speeding up

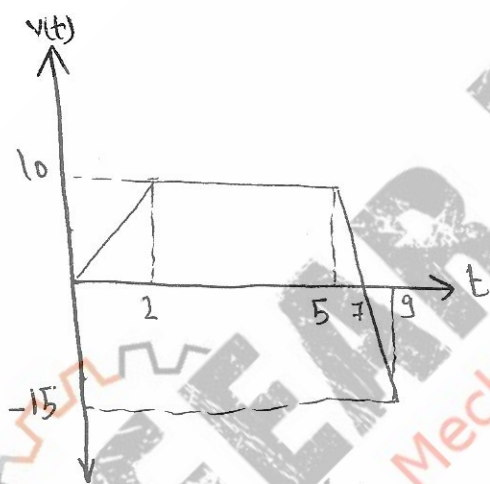
II) $v \rightarrow$
 $a \leftarrow$

The particle is speeding down

2.6) The particle under constant acceleration:

$$\begin{aligned} V_F &= V_i + at \\ (V_F)^2 &= (V_i)^2 + 2a(x_F - x_i) \\ X_F &= x_i + V_i t + \frac{1}{2} at^2 \end{aligned}$$

* Example:



a) Displacement from $t=0$ to $t=2$?

$$\text{area} = \frac{1}{2} \times 2 \times 10 = 10 \text{ m}$$

b) Displacement from $t=0$ to $t=9$?

Negative VP Because It is in the negative y-axis

$$\text{area} = 10 + 3 \times 10 + \frac{1}{2} \times 2 \times 10 - \left(\frac{1}{2} \times 2 \times 10 \right) = 35 \text{ m}$$

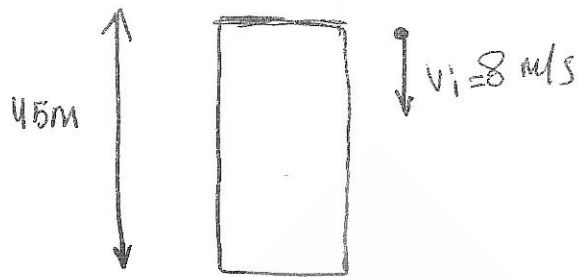
2.7) Freely Falling objects:

A Freely falling object is any object moving freely under the influence of gravity alone.

$$\vec{a} = -g = -9.8 \text{ m/s}^2$$

③ Example:

Find the time required to
reach the earth?
surface of the



—: solution:—

$$y_f = y_i + v_i t + \frac{1}{2} g t^2$$

$$0 = 45 - 8 \times t + \frac{1}{2} \times (-9.8) t^2$$

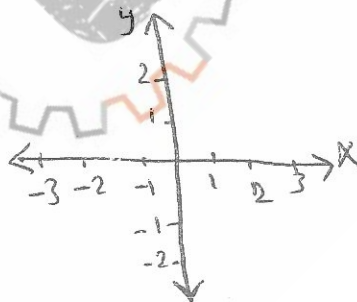
$$t = +8 \pm \frac{\sqrt{(8)^2 - (4)(45)(-4.9)}}{2 \times -4.9} \Rightarrow t = 2.32 \text{ s}$$

3.1

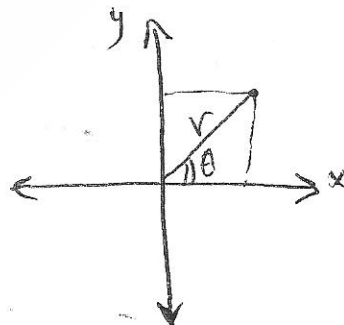
Chapter "3"

* coordinate systems:

Cartesian cord system



polar cord system



$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\tan \theta = \frac{y}{x}$$

$$r = \sqrt{x^2 + y^2}$$

3.2) vector and scalar magnitudes:

scalar: magnitude + ~~scalar~~ Unit $\Rightarrow$ mass, speed, distance

vector: magnitude + direction $\Rightarrow$ displacement, velocity

$$\text{magnitude of a vector } \vec{A} = |\vec{A}| = \sqrt{x^2 + y^2}$$

3.3) some properties of vectors:

I) Equality of two vectors:

$$\vec{A} = \vec{B} ; \begin{cases} |\vec{A}| = |\vec{B}| \\ \vec{A} \parallel \vec{B} \end{cases}$$

II) adding vectors:

$$\vec{A} + \vec{B} = \vec{B} + \vec{A} \quad [\text{commutative}]$$

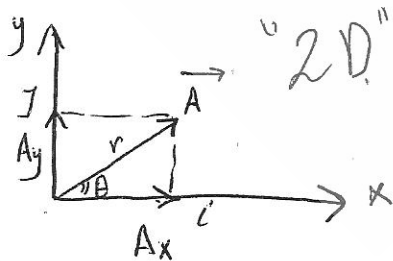
$$(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C}) \quad [\text{associative}]$$

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B}) \quad [\text{subtracting vectors}]$$

NOTE: $\vec{B}$, $-\vec{B}$ They have the same magnitude But opposite direction

$$k(\vec{A} + \vec{B}) = k\vec{A} + k\vec{B} \quad [\text{multiplying by a scalar}]$$

3.4) components of a vector and unit vectors:



$$\begin{cases} A_x = |\vec{A}| \cos \theta \\ A_y = |\vec{A}| \sin \theta \\ \vec{A} = A_x \hat{i} + A_y \hat{j} \end{cases}$$

$(\hat{i}, \hat{j}, \hat{k})$: Unit vectors magnitude "1"

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

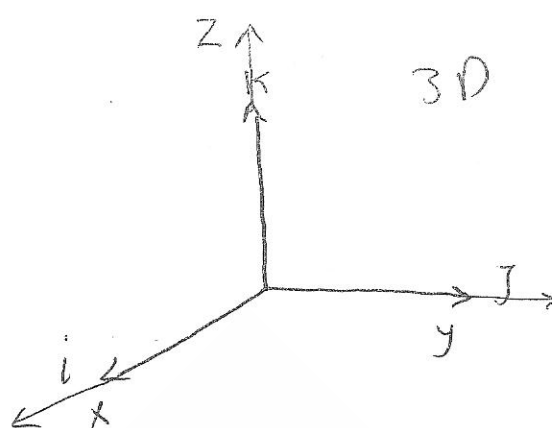
$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{R} = \vec{A} + \vec{B} =$$

$$(A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} + (A_z + B_z)\hat{k}$$

$$= R_x \hat{i} + R_y \hat{j} + R_z \hat{k}$$

$$|\vec{R}| = \sqrt{(R_x)^2 + (R_y)^2 + (R_z)^2}$$



Θ : Angle that $\vec{R}$ makes with

- x-axis $\rightarrow \theta_x = \cos^{-1} \frac{R_x}{|\vec{R}|}$
- y-axis $\rightarrow \theta_y = \cos^{-1} \frac{R_y}{|\vec{R}|}$
- z-axis $\rightarrow \theta_z = \cos^{-1} \frac{R_z}{|\vec{R}|}$

The scalar (dot) product of two vectors

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta$$

θ : Angle between the two vectors

$$\hat{i} \cdot \hat{i} = 1$$

$$\hat{j} \cdot \hat{j} = 1$$

$$\hat{k} \cdot \hat{k} = 1$$

$$\hat{i} \cdot \hat{j} = 0$$

$$\hat{i} \cdot \hat{k} = 0$$

$$\hat{j} \cdot \hat{k} = 0$$

$$\vec{A} \cdot \vec{A} = A_x^2 + A_y^2 + A_z^2 = |\vec{A}|^2$$

$$\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$$

The cross product of two vectors

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

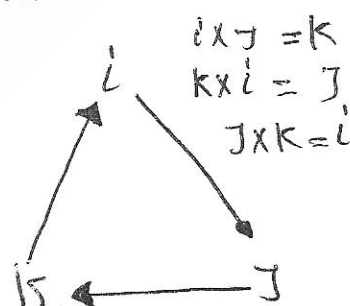
$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{A} \times \vec{B} = \vec{C}$$

$$\vec{C} \perp \vec{A}, \vec{B}$$

$$|\vec{A} \times \vec{B}| = |\vec{A}| |\vec{B}| \sin \theta = |\vec{C}|$$

θ : Angle between the vectors



$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$= (A_y B_z - A_z B_y) \hat{i} - (A_x B_z - A_z B_x) \hat{j} + (A_x B_y - A_y B_x) \hat{k}$$

$$* \vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

$$\vec{A} \times \vec{A} = 0$$

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

4.1):

"Chapter 4"

* Motion in 2-Dimensions: "position-velocity-acceleration":

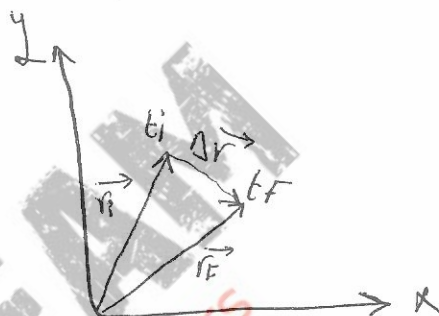
$$\Delta \vec{r} = \vec{r}_f - \vec{r}_i$$

$$\vec{v}_{avg} = \frac{\Delta \vec{r}}{\Delta t} \quad [\text{average velocity}]$$

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} \quad [\text{instantaneous velocity}]$$

$$\vec{a}_{avg} = \frac{\Delta \vec{v}}{\Delta t} \quad [\text{average acceleration}]$$

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} \quad [\text{instantaneous acceleration}]$$



4.2):

* Two Dimensional Motion with constant acceleration:

$$v_{fx} \hat{i} + v_{fy} \hat{j} = (v_{ix} + a_x t) \hat{i} + (v_{iy} + a_y t) \hat{j}$$

$$(v_{fx} \hat{i} + v_{fy} \hat{j})^2 = (v_{ix} \hat{i} + v_{iy} \hat{j})^2 + 2(a_x \hat{i} + a_y \hat{j}) \Delta x$$

$$(x_{fx} \hat{i} + x_{fy} \hat{j}) = (x_{ix} \hat{i} + x_{iy} \hat{j}) + (v_{ix} \hat{i} + v_{iy} \hat{j})t + \frac{1}{2}(a_x \hat{i} + a_y \hat{j})t^2$$

* Example: A particle starts from the origin at $t=0$, with an initial velocity having an x-component of 20 m/s and a y-component of -15 m/s. The particle moves in the x-y plane with an acceleration in the x-direction of 4 m/s²

- Find total velocity as a function of Time
- Find the velocity and speed at $t=5s$
- Find x-y components position

Solution: $x_i = 0$, $y_i = 0$, $v_{xi} = 20 \text{ m/s}$, $v_{yi} = -15 \text{ m/s}$, $a_x = 4 \text{ m/s}^2$, $a_y = 0$

$$\rightarrow a) \quad v_{xF} = v_{xi} + a_x t = 20 + 4t$$

$$v_{yF} = v_{yi} + a_y t = -15 + 0 = -15$$

$$\vec{v}_F = v_{xF} + v_{yF} = (20 + 4t)\hat{i} - 15\hat{j} \text{ m/s}$$

$$\rightarrow b) \quad \vec{v}(5s) = (20 + 4 \times 5)\hat{i} - 15\hat{j} = (40\hat{i} - 15\hat{j}) \text{ m/s}$$

$$\text{speed} = |\vec{v}| = \sqrt{(40)^2 + (-15)^2} = 43 \text{ m/s}$$

$$\rightarrow c) \quad x_F = x_i + v_{xi}t + \frac{1}{2}a_x t^2 = 0 + 20t + 2t^2 = (20t + 2t^2) \text{ m}$$

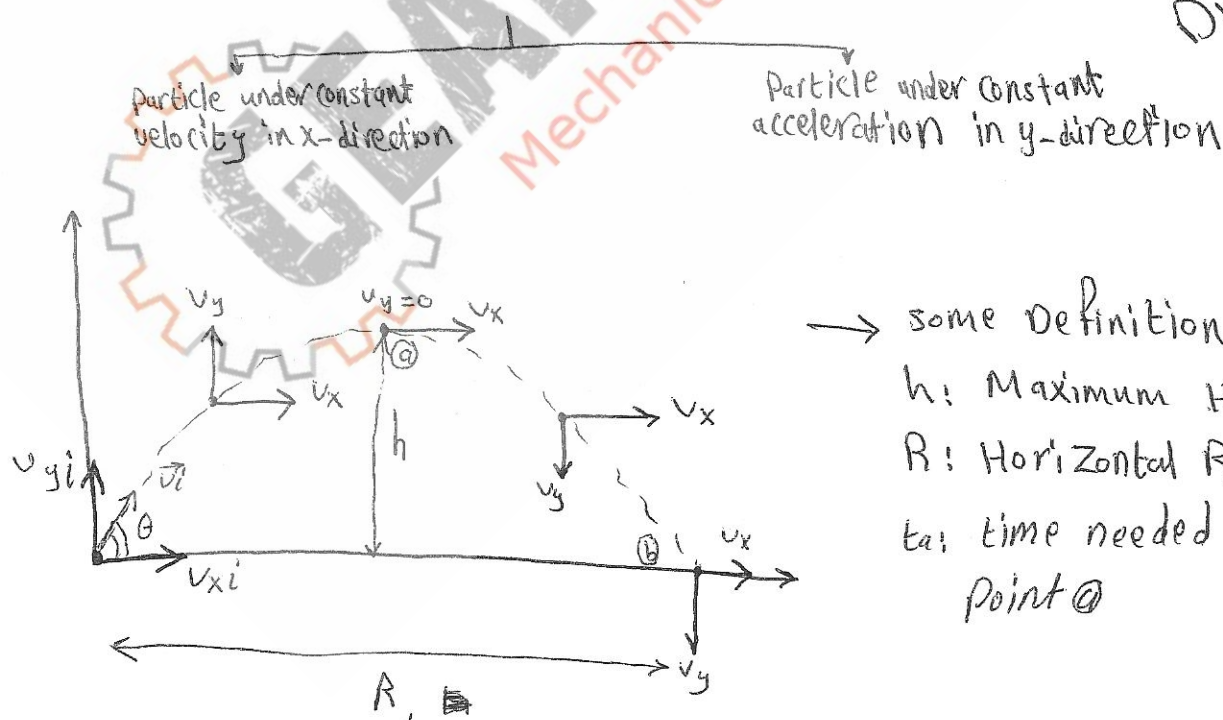
$$y_F = y_i + v_{yi}t + \frac{1}{2}a_y t^2 = 0 + (-15)t + 0 = (-15t) \text{ m}$$

$$\vec{r}_F = x_F \hat{i} + y_F \hat{j} = (20t + 2t^2)\hat{i} - 15t\hat{j} \text{ m}$$

4.3) projectile motion:

Dimensional motion, under constant

acceleration $\Rightarrow a_x = 0$, $a_y = -9.8 \text{ m/s}^2$



→ some definitions:

h : Maximum Height

R : Horizontal Range

t_a : time needed to reach point @

$$h = \frac{v_i^2 \sin^2 \theta_i}{2g}$$

$$R = \frac{v_i^2 \sin 2\theta_i}{g}$$

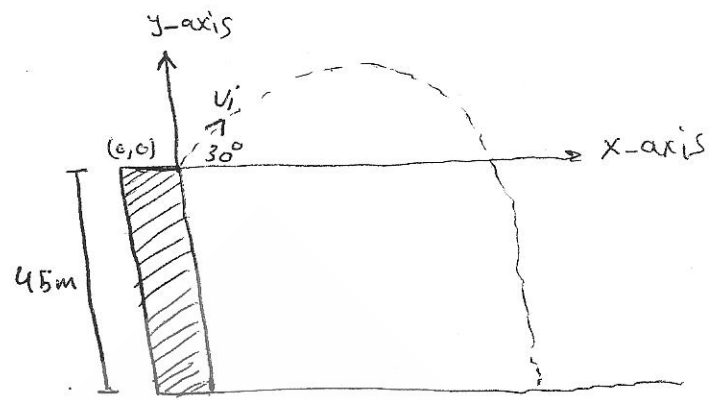
$$R_{\max} : \theta = 45^\circ$$

$$t_a = \frac{v_i \sin \theta_i}{g}$$

Be attention
NOT to add them
as they are in
different
Directions

* Example: A stone is thrown from top of a building upward at an angle of 30° above horizontal with initial speed 20 m/s

- a) How long does it take the stone to reach the ground?
 b) what is the speed of the stone just before it strikes the ground?



— Solution —

(a)

$$x_i = y_i = 0, \quad y_f = -45 \text{ m}$$

$$v_i = 20 \text{ m/s}, \quad \theta = 30^\circ$$

$$v_{ix} = v_i \cos \theta = 20 \times \frac{\sqrt{3}}{2} = 10\sqrt{3} \text{ m/s}$$

$$v_{iy} = v_i \sin \theta = 20 \times \frac{1}{2} = 10 \text{ m/s}$$

$$y_f = y_i + v_{iy}t + \frac{1}{2} a_y t^2 \Rightarrow -45 = 0 + 10 \times t + \frac{1}{2} \times (-9.8) \times t^2$$

$$\therefore 4.9t^2 - 10t - 45 = 0$$

$$\rightarrow t = 4.225$$

(b) $v_{xf} = v_{xi} = 17.3 \text{ m/s}$

$$v_{yf} = v_{yi} + a_y t = 10 - 9.8 \times 4.22 = -31.3 \text{ m/s}$$

$$\vec{v}_f = (17.3\hat{i} - 31.3\hat{j}) \text{ m/s} \quad \text{"velocity"}$$

$$|\vec{v}_f| = \sqrt{(17.3)^2 + (31.3)^2} = 35.8 \text{ m/s} \quad \text{"speed"}$$

4.4) particle in uniform circular motion:

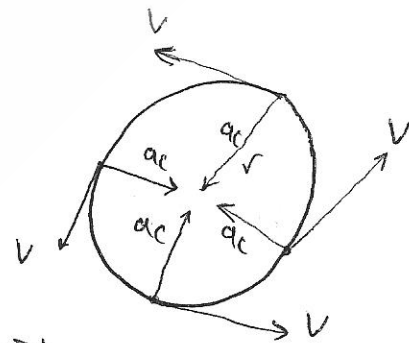
A particle moving in a circle of radius r , with constant speed v , is uniform circular motion.

* Some Definitions:

Radial acceleration ($\vec{a}_r$)

$$\vec{a}_r \rightarrow |\vec{a}_r| = |\vec{a}_c|$$

Centripetal acceleration ($\vec{a}_c$)



$$|\vec{a}_c| = \frac{v^2}{r}, \quad T = \frac{2\pi r}{v}$$

$$\vec{a}_c \perp \vec{v}$$

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4.5) Tangential and radial acceleration:

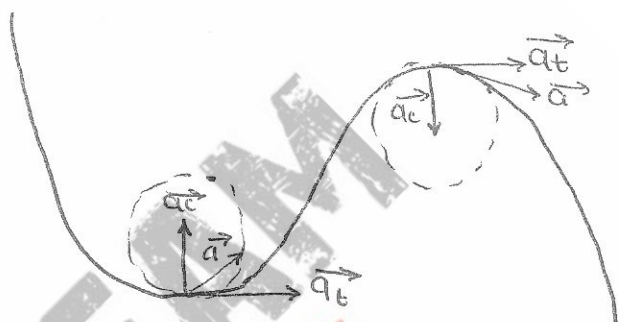
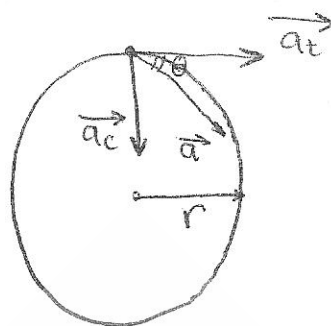
$$\rightarrow \vec{a}_t \perp \vec{a}_c, \vec{a}_r$$

$$\rightarrow |\vec{a}| = \sqrt{(a_c)^2 + (a_t)^2} \quad [\text{magnitude of total acceleration}]$$

$$\rightarrow \theta = \tan^{-1} \left(\frac{a_c}{a_t} \right)$$

$$\rightarrow |\vec{a}_c| = \frac{v^2}{r}$$

$$\rightarrow |\vec{a}_t| = \left| \frac{dv}{dt} \right| \quad \begin{cases} \text{increasing} \rightarrow (+v) \\ \text{decreasing} \rightarrow (-v) \end{cases}$$



"Chapter 5"

5.1) The concept of Force:

$$\text{Force (vector)} = \vec{F}, \quad \text{Unit: Newton [N]}$$

5.2) Newton's First law:

In the absence of external forces and when viewed from an inertial reference frame an object at rest remains at rest, and an object in motion continues in motion with a constant velocity

5.3) Mass:

Mass: It is that property of an object that specifies how much resistance an object exhibits to changes in its velocity.

M: Unit Kg

$$\frac{W}{F} = \frac{Mg}{a} \rightarrow M \propto a$$

5.4) Newton's second law:

The acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass.

$$\sum \vec{F} = M \vec{a}$$

$\downarrow$ $\downarrow$
 net force Mass acceleration

$$\begin{cases} \sum F_x = Ma_x \\ \sum F_y = Ma_y \end{cases}$$

5.5) Gravitational Force and weight:

$$\vec{F}_g = m\vec{g} \rightarrow \text{Free fall acceleration}$$

$\downarrow$ $\downarrow$
 gravitational force mass

5.6) Newton's Third law:

If two objects interact, The force $\vec{F}_{12}$ exerted by object (1) on object (2) is equal in magnitude and opposite to the force $\vec{F}_{21}$ exerted by object (2) on object (1)

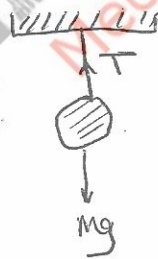
5.7) Applications:

I particle in equilibrium

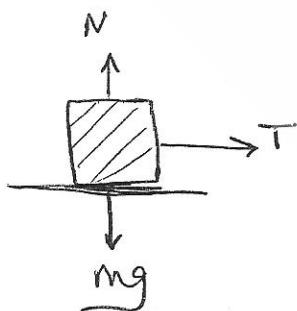
$$\sum \vec{F} = 0, \quad \vec{a} = 0$$

$$\sum F_y = 0 \rightarrow T - mg = 0$$

$$T = mg$$



II particle under a net force



$$\rightarrow \sum F_x = ma_x$$

$$\therefore T = ma_x$$

$$\rightarrow \sum F_y = 0$$

$$N = mg$$

* Example: A car of mass "M" is on an icy drive way inclined at angle θ
 a) Find acceleration

b) If the car is released from rest, how long does it take the car to reach the bottom of hill, and what is speed as it arrives there?!

→ Solution:

a) $\Sigma F_x = \text{Max}$

$mg \sin \theta = \text{Max} \rightarrow a_x = g \sin \theta$

b) $X_F = X_i + V_{xi}t + \frac{1}{2}a_x t^2$

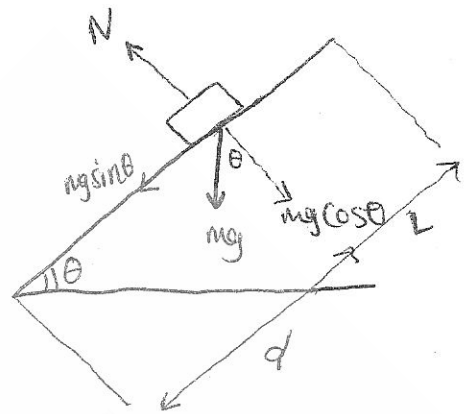
$d = 0 + 0 + \frac{1}{2}a_x t^2$

$t = \sqrt{\frac{2d}{a_x}} \rightarrow t = \sqrt{\frac{2d}{g \sin \theta}}$

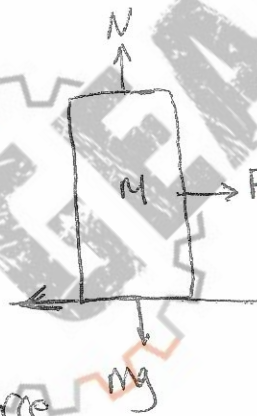
$V_{xF}^2 = V_{xi}^2 + 2a_x(X_F - X_i)$

$V_{xF}^2 = 0 + 2a_x d \Rightarrow$

$V_{xF} = \sqrt{2g \sin \theta d} = \sqrt{2dg \sin \theta}$



* 5.8) Forces of Friction:

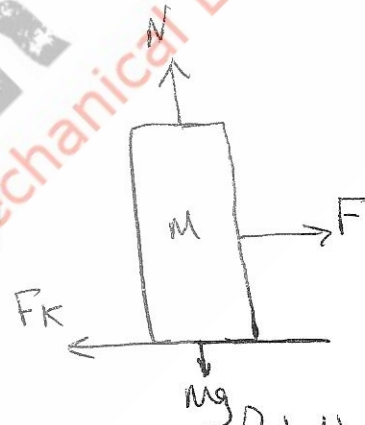


F_s : static Friction Force

$F_s \leq \mu_s N$

N: Normal Force

μ_s : coefficient of static friction

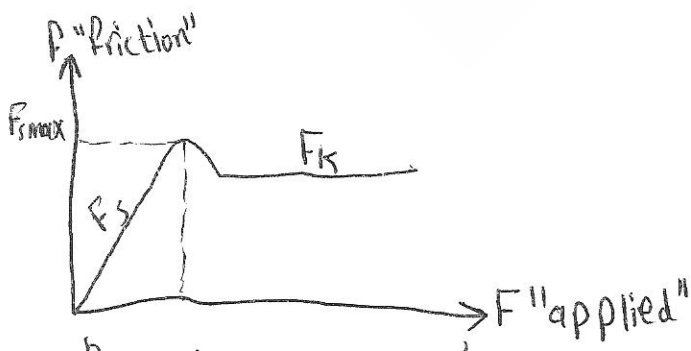


F_k : kinetic Friction Force

$F_k \leq \mu_k N$

N: Normal Force

μ_k : coefficient of kinetic friction



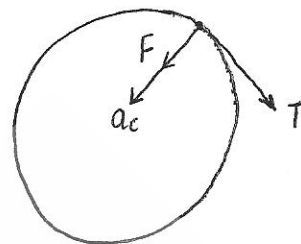
"Chapter 6"

6.1) Newton's second law for a particle in uniform circular motion:

Constant speed (v) $\Rightarrow$

$$a_c = \frac{v^2}{r}, \quad a_t = 0, \quad a_c \perp v$$

$$\Sigma F = ma_c = \frac{mv^2}{r}$$



* Example: A small ball of mass m is suspended from a string of length L . The ball revolves with constant speed v in a horizontal circle of radius r . Find an expression for v

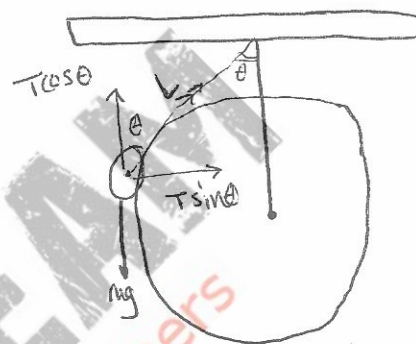
Solution:

$$\Sigma F_y = 0$$

$$T \cos \theta = mg \rightarrow T = \frac{mg}{\cos \theta} \quad \text{I}$$

$$\Sigma F_x = ma_c \rightarrow \frac{T \sin \theta}{\sin \theta} = \frac{mv^2}{r \sin \theta} \quad \text{II}$$

$$\frac{mg}{\cos \theta} = \frac{mv^2}{r \sin \theta} \quad \therefore v = \sqrt{rg \tan \theta}$$



$$\sin \theta = \frac{r}{L} \quad \therefore r = L \sin \theta \quad \therefore v = \sqrt{L g \sin \theta \tan \theta}$$

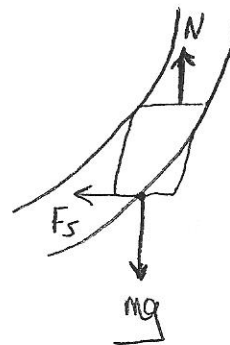
* Example: A 1500 kg car, moving on a flat horizontal road. If the radius of the curve of the road is 35 m and $\mu_s = 0.523$. Find maximum speed the car can have and still make the turn successfully

Solution:

$$F_{s \max} = m \frac{v_{\max}^2}{r}$$

$$\mu_s \cdot N = \frac{m v_{\max}^2}{r} \rightarrow v_{\max} = \sqrt{\mu_s \cdot g \cdot r}$$

$$v_{\max} = \sqrt{(0.523)(9.8)(35)} = 13.4 \text{ m/s}$$

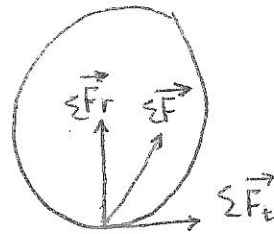


6.2) Non-uniform circular motion

$\Sigma \vec{F}_r \rightarrow$ Centre

$\Sigma \vec{F}_t \rightarrow$ Tangent to the circle

Varying speed: $\vec{a} = \vec{a}_r + \vec{a}_t$



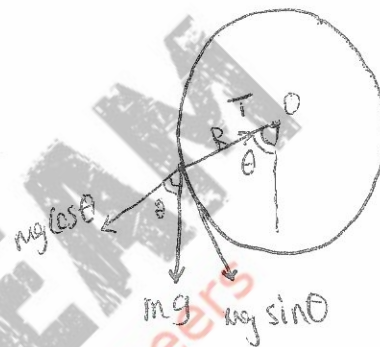
*Example: A small sphere of mass $[M]$ is attached to the end of a cord of length R , and set into motion in a vertical circle about a fixed point O . Find the tension in the cord at any instant when the speed of the sphere is v and the cord makes an angle θ with the vertical.

$$\Sigma F = ma_t \Rightarrow mg \sin \theta = ma_t$$

$$a_t = g \sin \theta$$

$$\Sigma F_r = ma_c \Rightarrow T - mg \cos \theta = \frac{mv^2}{R}$$

$$T = mg \left(\frac{v^2}{Rg} + \cos \theta \right)$$



Total acceleration:

$$|\vec{a}| = \sqrt{(|a_r|)^2 + (|a_t|)^2}$$

$$T = \begin{cases} \text{bottom: } (\theta = 0) : mg \left(\frac{v_{bot}^2}{Rg} + 1 \right) \\ \text{top: } (\theta = 180) : mg \left(-\frac{v_{top}^2}{Rg} - 1 \right) \end{cases}$$

مع الشكر والتقدير
للزميل

مهند اللحام

هندسة ميكاترونيكس

